17	**B**	Narrow tube ⇒ volume of tube is negligible. Steady state ⇒ the pressure in both containers is the same. Using $pV = nRT$, we have $pV = (m'/M)RT$ where m' and M represent the mass of the gas and molar mass of the gas respectively. Thus, $\frac{m'_X RT_X}{MV_X} = \frac{m'_Y RT_Y}{MV_Y}$ $\frac{m(300)}{2V} = \frac{m'_Y(400)}{V}$ $m'_Y = \frac{3}{8}m$
18	**D**	When heated, heat flows into the system making Q positive. The gas expands meaning W